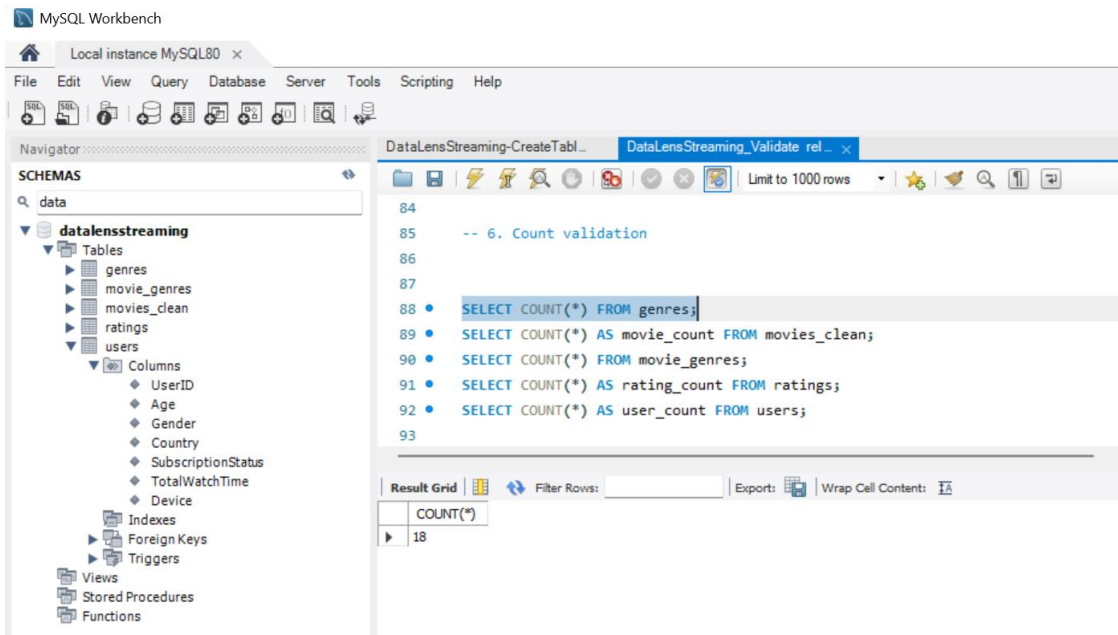


Task 6: Implement User Table in SQL

Validate relationships

1. Count validation

I. `SELECT COUNT(*) FROM genres;`



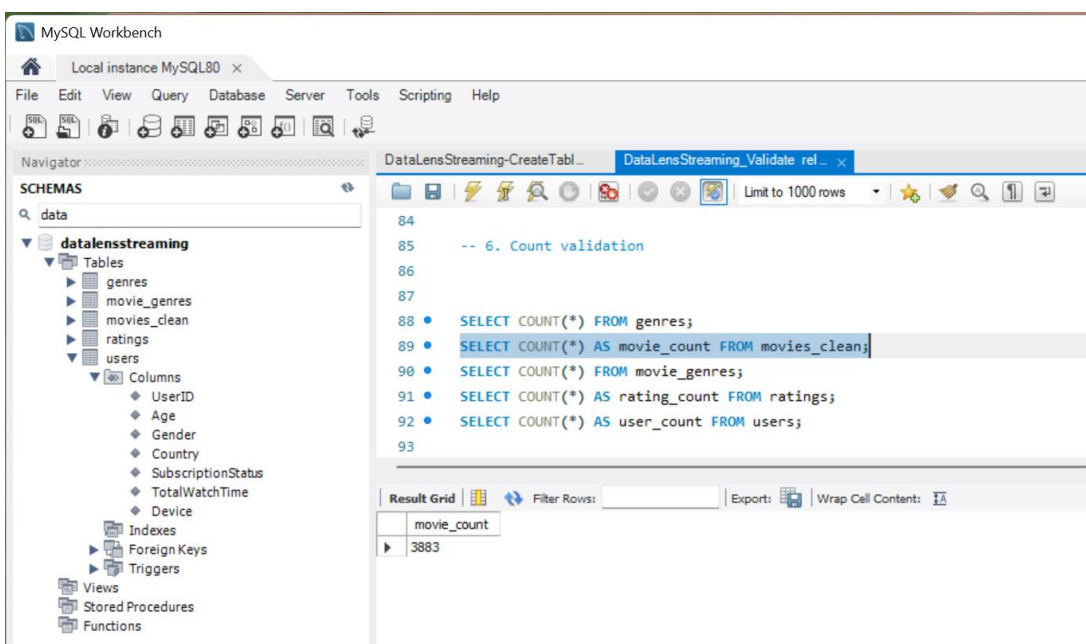
The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane shows the 'data' database selected, with the 'datalensstreaming' schema expanded. The 'Tables' list includes 'genres', 'movie_genres', 'movies_clean', 'ratings', and 'users'. The 'users' table is expanded, showing columns: UserID, Age, Gender, Country, SubscriptionStatus, TotalWatchTime, and Device. The 'Indexes' pane shows 'Foreign Keys' and 'Triggers'. The 'Query' pane on the right shows a SQL script with the following queries:

```
-- 6. Count validation
SELECT COUNT(*) FROM genres;
SELECT COUNT(*) AS movie_count FROM movies_clean;
SELECT COUNT(*) FROM movie_genres;
SELECT COUNT(*) AS rating_count FROM ratings;
SELECT COUNT(*) AS user_count FROM users;
```

The 'Result Grid' at the bottom shows the result of the first query:

COUNT(*)
18

II. `SELECT COUNT(*) AS movie_count FROM movies_clean;`



The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane shows the 'data' database selected, with the 'datalensstreaming' schema expanded. The 'Tables' list includes 'genres', 'movie_genres', 'movies_clean', 'ratings', and 'users'. The 'users' table is expanded, showing columns: UserID, Age, Gender, Country, SubscriptionStatus, TotalWatchTime, and Device. The 'Indexes' pane shows 'Foreign Keys' and 'Triggers'. The 'Query' pane on the right shows a SQL script with the following queries:

```
-- 6. Count validation
SELECT COUNT(*) FROM genres;
SELECT COUNT(*) AS movie_count FROM movies_clean;
SELECT COUNT(*) FROM movie_genres;
SELECT COUNT(*) AS rating_count FROM ratings;
SELECT COUNT(*) AS user_count FROM users;
```

The 'Result Grid' at the bottom shows the result of the second query:

movie_count
3883

III. SELECT COUNT(*) FROM movie_genres;

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'data' selected. Under 'data', the 'datalensstreaming' database is expanded, showing tables: 'genres', 'movie_genres', 'movies_clean', 'ratings', and 'users'. The 'movie_genres' table is highlighted. The main editor window shows a SQL script with the following queries:

```
84
85 -- 6. Count validation
86
87
88 • SELECT COUNT(*) FROM genres;
89 • SELECT COUNT(*) AS movie_count FROM movies_clean;
90 • SELECT COUNT(*) FROM movie_genres;
91 • SELECT COUNT(*) AS rating_count FROM ratings;
92 • SELECT COUNT(*) AS user_count FROM users;
93
```

The 'Result Grid' at the bottom shows the results of the selected query:

COUNT(*)
6404

IV. SELECT COUNT(*) AS rating_count FROM ratings;

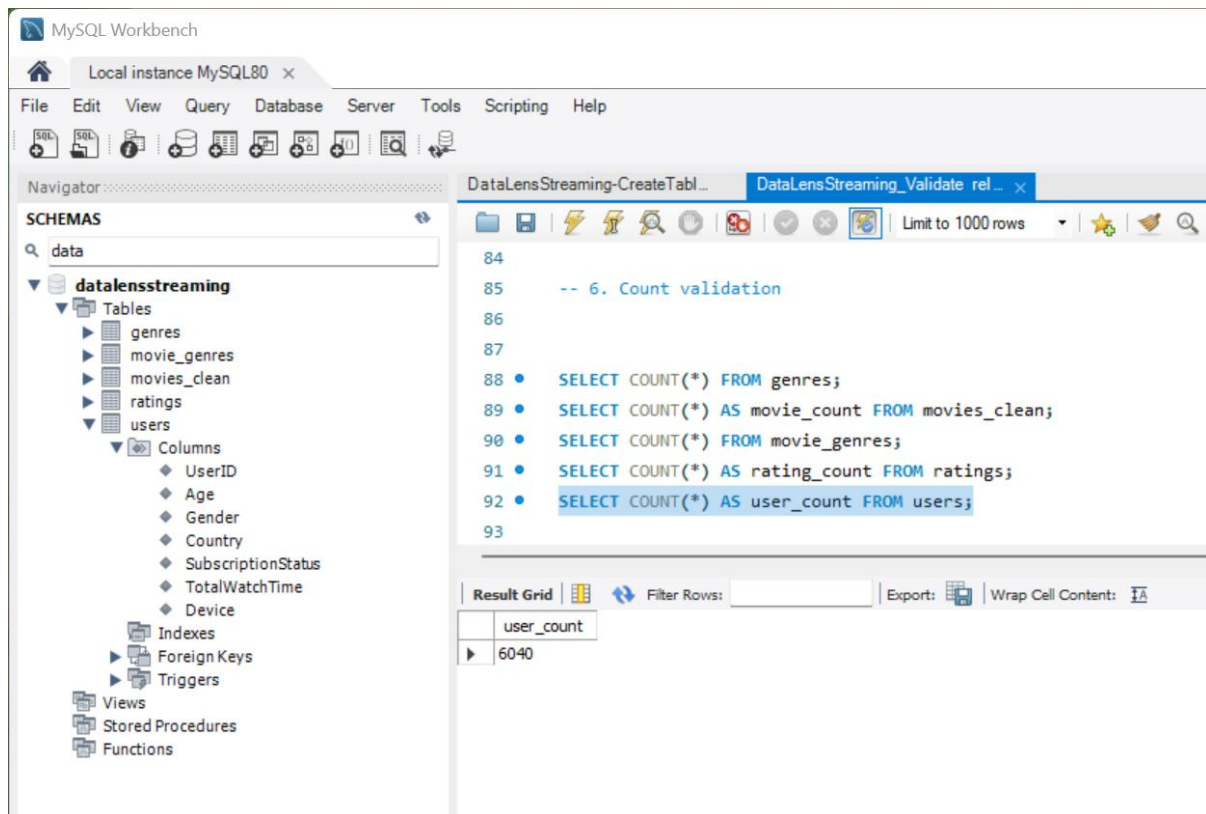
The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'data' selected. Under 'data', the 'datalensstreaming' database is expanded, showing tables: 'genres', 'movie_genres', 'movies_clean', 'ratings', and 'users'. The 'ratings' table is highlighted. The main editor window shows a SQL script with the following queries:

```
84
85 -- 6. Count validation
86
87
88 • SELECT COUNT(*) FROM genres;
89 • SELECT COUNT(*) AS movie_count FROM movies_clean;
90 • SELECT COUNT(*) FROM movie_genres;
91 • SELECT COUNT(*) AS rating_count FROM ratings;
92 • SELECT COUNT(*) AS user_count FROM users;
93
```

The 'Result Grid' at the bottom shows the results of the selected query:

rating_count
10000

V. SELECT COUNT(*) AS user_count FROM users



2. Validate Users → Ratings relationship

```
SELECT  
  
r.RatingID,  
  
r.UserID,  
  
u.Age,  
  
u.Gender,  
  
u.Country,  
  
u.SubscriptionStatus,  
  
u.TotalWatchTime,  
  
u.Device,  
  
r.MovieID,  
  
r.Rating,  
  
r.Timestamp  
FROM ratings r
```

JOIN users u

ON r.UserID = u.UserID

LIMIT 20;

The screenshot shows the MySQL Workbench interface. On the left, the 'Navigator' pane displays the 'datastreaming' schema with tables like 'genres', 'movie_genres', 'movies_clean', 'ratings', and 'users'. The 'users' table is selected, showing its columns: UserID, Age, Gender, Country, SubscriptionStatus, TotalWatchTime, Device, and Rating. The main editor shows a SQL query titled '1. Validate Users + Ratings relationship'. The query is a SELECT statement joining the 'ratings' table (r) and the 'users' table (u) on the condition r.UserID = u.UserID, limiting the results to 20 rows. The 'Result Grid' at the bottom displays the query results as a table with 27 rows and 10 columns: RatingID, UserID, Age, Gender, Country, SubscriptionStatus, TotalWatchTime, Device, MovieID, and Rating. The results show a list of ratings for various users, including their age, gender, country, subscription status, total watch time, device used, movie ID, and rating score.

RatingID	UserID	Age	Gender	Country	SubscriptionStatus	TotalWatchTime	Device	MovieID	Rating
1	SF_2741	50	M	India	Subscriber	395	Desktop	3713	4.1
2	SF_3082	41	F	Canada	Free	497	SmartTV	2052	3.1
3	SF_33	16	M	Italy	Free	54	Tablet	3267	4
4	SF_182	14	M	India	Free	236	SmartTV	2517	2.4
5	SF_631	20	M	Canada	Free	99	Mobile	2996	3.1
6	SF_2261	31	M	India	Free	256	SmartTV	1184	3.9
7	SF_1291	42	F	USA	Subscriber	187	Tablet	2425	2.8
9	SF_693	62	M	UK	Subscriber	325	SmartTV	1375	1.9
11	SF_1503	24	M	Canada	Free	337	Mobile	127	4.6
12	SF_1907	43	M	France	Free	176	Desktop	1138	1.2
15	SF_2147	11	F	USA	Subscriber	367	Desktop	2493	2.7
16	SF_2215	21	F	Italy	Free	21	Tablet	2912	2
18	SF_601	42	F	USA	Free	33	Desktop	2720	2.7
19	SF_841	19	F	USA	Subscriber	413	Mobile	2955	4.1
20	SF_1504	54	M	Italy	Subscriber	249	Tablet	2351	2.7
21	SF_3388	63	M	Italy	Subscriber	20	Mobile	2653	4.8
22	SF_1839	65	M	Italy	Subscriber	106	Mobile	1509	2
25	SF_363	35	M	Canada	Free	76	Mobile	22	2.4
26	SF_2617	30	F	India	Free	370	SmartTV	1029	3.7
27	SF_2919	11	M	USA	Subscriber	145	Tablet	1493	2.8

3. Validate Movies → Ratings relationship

SELECT

r.RatingID,

r.MovieID,

m.Movie_Title,

m.Year,

m.Language,

m.Country,

m.Total_Views,

r.UserID,

r.Rating,

```

r.Timestamp

FROM ratings r

JOIN movies_clean m

    ON r.MovieID = m.MovieID

LIMIT 20;

```

The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane displays the 'data' database with a tree view of tables including 'genres', 'movie_genres', 'movies_clean', 'ratings', and 'users'. The 'ratings' table is selected. The main editor shows a SQL query titled 'DataLensStreaming-Validate rel...'. The query is as follows:

```

-- 2. Validate Movies + Ratings relationship
SELECT
    r.RatingID,
    r.MovieID,
    m.Movie_Title,
    m.Year,
    m.Language,
    m.Country,
    m.Total_Views,
    r.UserID,
    r.Ratings,
    r.Timestamp
FROM ratings r
JOIN movies_clean m
    ON r.MovieID = m.MovieID
LIMIT 20;

```

Below the query editor, the 'Result Grid' shows the first 20 rows of the query results. The columns are RatingID, MovieID, Movie_Title, Year, Language, Country, Total_Views, UserID, Ratings, and Timestamp.

RatingID	MovieID	Movie_Title	Year	Language	Country	Total_Views	UserID	Ratings	Timestamp
1	3713	Long Walk Home, The	1990	French	Canada	86324	SF_2741	4.1	2024-02-01 04:32:50
2	2052	Hocus Pocus	1993	Spanish	India	4381	SF_3082	3.1	2024-02-24 04:32:50
3	3267	Meriachi, El	1992	Italian	Italy	11502	SF_53	4	2024-11-12 04:32:50
4	2517	Christine	1983	English	Italy	56059	SF_182	2.4	2024-09-28 04:32:50
5	2996	Music of the Heart	1999	Spanish	UK	11651	SF_631	3.1	2024-10-04 04:32:50
6	1184	Mediterraneo	1991	German	USA	89764	SF_2261	3.9	2024-05-28 04:32:50
7	2425	General, The	1998	German	UK	24747	SF_1291	2.8	2023-12-24 04:32:50
8	3326	What Planet Are You From?	2000	Italian	UK	97330	SF_5002	4.5	2024-03-05 04:32:50
9	1375	Star Trek III: The Search for Spock	1984	Italian	Italy	31440	SF_693	1.9	2024-12-05 04:32:50
10	1987	From Night	1980	Spanish	Italy	43433	SF_3684	3.2	2024-07-16 04:32:50
11	127	Silence of the Palace, The (Saint ...	1994	Italian	USA	72789	SF_1503	4.6	2024-01-22 04:32:50
12	1138	Daddystown	1995	German	Italy	97867	SF_1907	1.2	2024-02-04 04:32:50
13	734	Getting Away With Murder	1996	Spanish	France	89730	SF_3649	4.6	2024-11-26 04:32:50
14	309	Red Firecracker, Green Firecracker	1994	German	France	72056	SF_4824	2.4	2024-02-24 04:32:50
15	2493	Harmonists, The	1997	French	France	65671	SF_2147	2.7	2024-09-21 04:32:50
16	2912	Limy, The	1999	German	India	67268	SF_2215	2	2024-04-23 04:32:50
17	2983	Icpress File, The	1965	English	India	60376	SF_6005	2.8	2024-04-11 04:32:50
18	2720	Inspector Gadget	1999	Italian	India	97145	SF_601	2.7	2024-07-11 04:32:50
19	2955	Penitentiary II	1982	German	India	92195	SF_841	4.1	2024-08-25 04:32:50
20	2351	Nights of Cabiria (Le Notte di Cabiria)	1957	Spanish	India	52296	SF_1504	2.7	2024-12-03 04:32:50

4. Validate Users + Movies + Ratings together

```

SELECT

r.RatingID,

u.UserID,

u.Age,

u.Gender,

u.Country AS UserCountry,

```

```

u.SubscriptionStatus,

m.MovieID,

m.Movie_Title,

m.Country AS MovieCountry,

m.Language,

m.Year,

r.Rating,

r.Timestamp

FROM ratings r

JOIN users u

    ON r.UserID = u.UserID

JOIN movies_clean m

    ON r.MovieID = m.MovieID

LIMIT 20;

```

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'data' selected. The main editor shows a SQL query titled 'DataLensStreaming-CreateTabl...' and 'DataLensStreaming-Validate rel...'. The query is a SELECT statement joining 'ratings', 'users', and 'movies_clean' tables, limited to 20 rows. The bottom panel shows the 'Result Grid' with columns: RatingID, MovieID, Movie_Title, Year, Language, Country, Total_Views, UserID, Rating, and Timestamp. The results list 20 rows of data.

RatingID	MovieID	Movie_Title	Year	Language	Country	Total_Views	UserID	Rating	Timestamp
1	3713	Long Walk Home, The	1990	French	Canada	86324	SF_2741	4.1	2024-02-01 04:32:50
2	2052	Hocus Pocus	1993	Spanish	India	4381	SF_3082	3.1	2024-02-24 04:32:50
3	3267	Marachi, El	1992	Italian	Italy	11502	SF_53	4	2024-11-12 04:32:50
4	2517	Christine	1983	English	Italy	56059	SF_182	2.4	2024-09-28 04:32:50
5	2996	Music of the Heart	1999	Spanish	UK	11651	SF_631	3.1	2024-10-04 04:32:50
6	1184	Mediterraneo	1991	German	USA	89764	SF_2261	3.9	2024-05-28 04:32:50
7	2425	General, The	1998	German	UK	24747	SF_1291	2.8	2023-12-24 04:32:50
8	3326	What Planet Are You From?	2000	Italian	UK	97330	SF_5002	4.5	2024-03-05 04:32:50
9	1375	Star Trek III: The Search for Spock	1984	Italian	Italy	31440	SF_693	1.9	2024-12-05 04:32:50
10	1987	Prom Night	1980	Spanish	Italy	43433	SF_3684	3.2	2024-07-16 04:32:50
11	127	Silence of the Palace, The (Saunt ...	1994	Italian	USA	72789	SF_1503	4.6	2024-01-22 04:32:50
12	1138	Dadetown	1995	German	Italy	97867	SF_1907	1.2	2024-02-04 04:32:50
13	724	Getting Away With Murder	1996	Spanish	France	89730	SF_3649	4.6	2024-11-26 04:32:50
14	309	Red Firecracker, Green Firecracker	1994	German	France	72055	SF_4824	2.4	2024-02-24 04:32:50
15	2493	Harmonists, The	1997	French	France	65671	SF_2147	2.7	2024-09-21 04:32:50
16	2912	Limey, The	1999	German	India	67268	SF_2215	2	2024-04-23 04:32:50
17	2983	Ipocress File, The	1965	English	India	60376	SF_6005	2.8	2024-04-11 04:32:50
18	2720	Inspector Gadget	1999	Italian	India	97145	SF_601	2.7	2024-07-11 04:32:50
19	2955	Penitentiary II	1982	German	India	92195	SF_841	4.1	2024-08-25 04:32:50
20	2351	Nights of Cabiria (Le Notti di Cabiria)	1957	Spanish	India	52296	SF_1504	2.7	2024-12-03 04:32:50

5. Check orphan UserID records in ratings

This checks whether any rating belongs to a user that does not exist in users.

```
SELECT

    r.UserID,

    COUNT(*) AS orphan_rating_count

FROM ratings r

LEFT JOIN users u

    ON r.UserID = u.UserID

WHERE u.UserID IS NULL

GROUP BY r.UserID;
```

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'data' selected. The main editor shows a SQL query with line numbers 59 to 71. The query is identical to the one in the previous block. Below the query editor, the 'Result Grid' is visible, showing a table with 11 columns: RatingID, UserID, Age, Gender, UserCountry, SubscriptionStatus, MovieID, MovieTitle, MovieCountry, Language, Year, Rating, and Timestamp. The table contains 25 rows of data, starting with RatingID 1 and ending with 25. The data includes various user details and movie ratings.

RatingID	UserID	Age	Gender	UserCountry	SubscriptionStatus	MovieID	MovieTitle	MovieCountry	Language	Year	Rating	Timestamp
1	SF_2741	50	M	India	Subscriber	3713	Long Walk Home, The	Canada	French	1990	4.1	2024-02-01 04:32:50
2	SF_3062	41	F	Canada	Free	2052	House Pocus	India	Spanish	1993	3.1	2024-02-24 04:32:50
3	SF_33	16	M	Italy	Free	3267	Marachi, El	Italy	Italian	1992	4	2024-11-12 04:32:50
4	SF_182	14	M	India	Free	2517	Christine	Italy	English	1983	2.4	2024-09-28 04:32:50
5	SF_631	20	M	Canada	Free	2996	Music of the Heart	UK	Spanish	1999	3.1	2024-10-04 04:32:50
6	SF_2261	31	M	India	Free	1184	Mediterraneo	USA	German	1991	3.9	2024-05-28 04:32:50
7	SF_1291	42	F	USA	Subscriber	2425	General, The	UK	German	1998	2.8	2023-12-24 04:32:50
9	SF_693	62	M	UK	Subscriber	1375	Star Trek III: The Search for Spock	Italy	Italian	1984	1.9	2024-12-05 04:32:50
10	SF_3684	42	M	France	Free	1967	Prom Night	Italy	Spanish	1980	3.2	2024-07-16 04:32:50
11	SF_1503	24	M	Canada	Free	127	Silence of the Palace, The (Saint el Qusur)	USA	Italian	1994	4.6	2024-01-22 04:32:50
12	SF_1907	43	M	France	Free	1138	Dadetown	Italy	German	1995	1.2	2024-02-04 04:32:50
13	SF_3649	48	F	Canada	Free	734	Getting Away With Murder	France	Spanish	1996	4.6	2024-11-26 04:32:50
15	SF_2147	11	F	USA	Subscriber	2493	Harmonists, The	France	French	1997	2.7	2024-09-21 04:32:50
16	SF_2215	21	F	Italy	Free	2912	Imrey, The	India	German	1999	2	2024-04-23 04:32:50
18	SF_601	42	F	USA	Free	2720	Inspector Gadget	India	Italian	1999	2.7	2024-07-11 04:32:50
19	SF_841	19	F	USA	Subscriber	2955	Penitentiary II	India	German	1982	4.1	2024-08-25 04:32:50
20	SF_1504	54	M	Italy	Subscriber	2351	Nights of Cabiria (Le Notte di Cabiria)	India	Spanish	1957	2.7	2024-12-03 04:32:50
21	SF_3288	63	M	Italy	Subscriber	2853	Communism (a.k.a. Alice, Sweet Alice)/Hol...	India	French	1977	4.8	2024-02-06 04:32:50
22	SF_1839	65	M	Italy	Subscriber	1509	All Over Me	USA	Spanish	1997	2	2024-06-28 04:32:50
25	SF_363	35	M	Canada	Free	22	Copycat	France	Italian	1995	2.4	2024-08-07 04:32:50

6. Check orphan MovieID records in ratings

This checks whether any rating belongs to a movie that does not exist in movies_clean.

```
SELECT

    r.MovieID,

    COUNT(*) AS orphan_rating_count

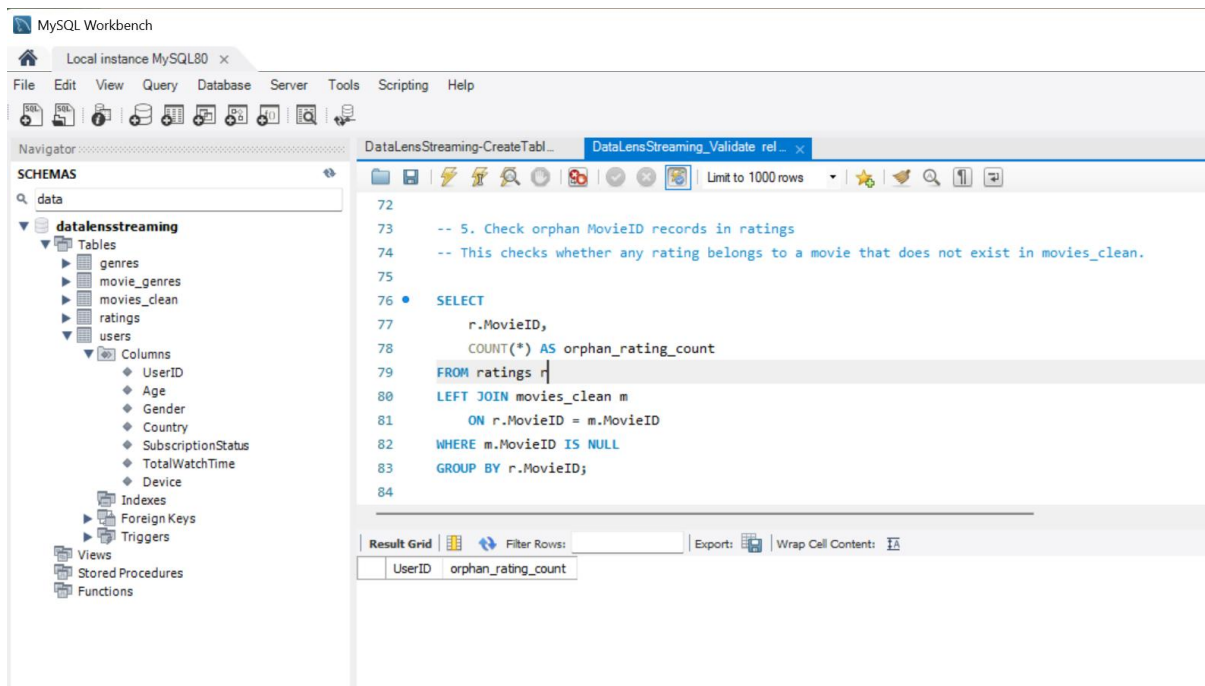
FROM ratings r

LEFT JOIN movies_clean m

    ON r.MovieID = m.MovieID

WHERE m.MovieID IS NULL

GROUP BY r.MovieID;
```

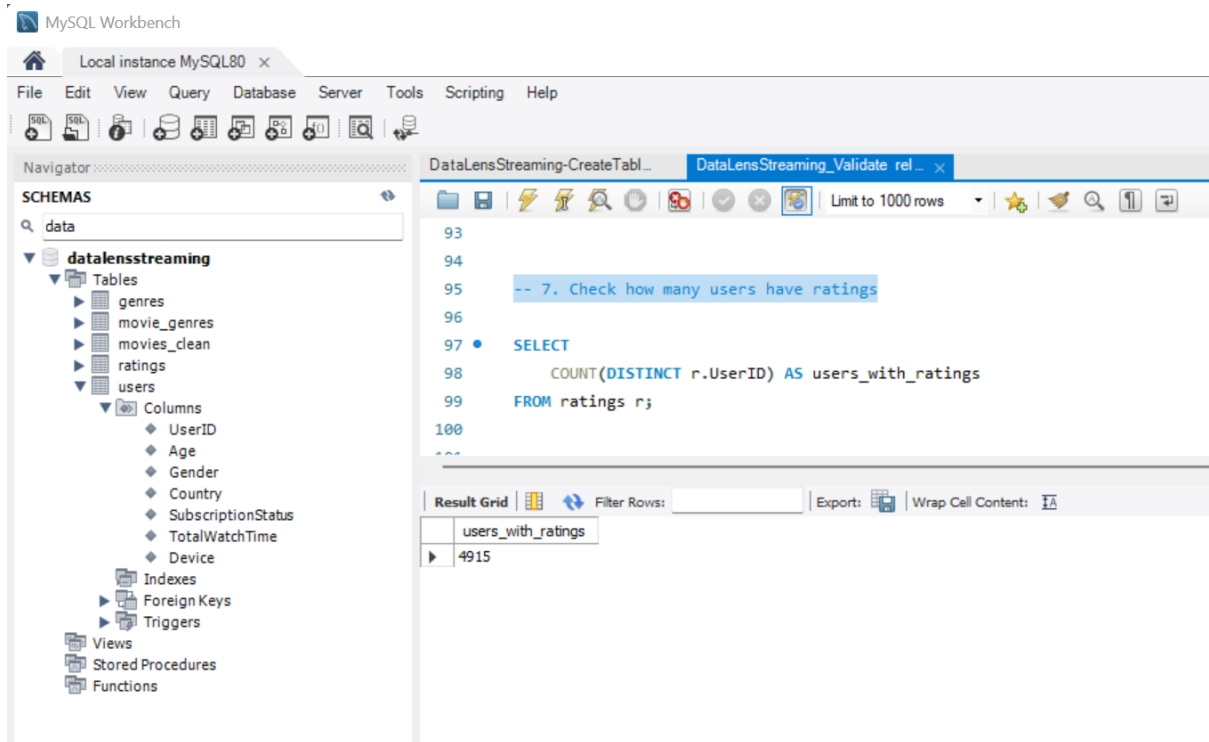


7. Check how many users have ratings

```
SELECT

    COUNT(DISTINCT r.UserID) AS users_with_ratings

FROM ratings r;
```

8. Check users without ratings

```
SELECT
  u.UserID,
  u.Age,
  u.Gender,
  u.Country,
  u.SubscriptionStatus,
  u.TotalWatchTime,
  u.Device
FROM users u
LEFT JOIN ratings r
  ON u.UserID = r.UserID
WHERE r.UserID IS NULL;
```

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: data

SCHEMAS

- data
- dataensstreaming
 - Tables
 - genres
 - movie_genres
 - movies_clean
 - ratings
 - users
 - Columns
 - UserID
 - Age
 - Gender
 - Country
 - SubscriptionStatus
 - TotalWatchTime
 - Device
 - Indexes
 - Foreign Keys
 - Triggers
 - Views
 - Stored Procedures
 - Functions

Showing loaded schemas only

Administration Schemas

Information

No object selected

DataLensStreaming-CreateTabl... DataLensStreaming_Validate rel... x

Limit to 1000 rows

```
101
102 -- 8. Check users without ratings
103
104 • SELECT
105     u.UserID,
106     u.Age,
107     u.Gender,
108     u.Country,
109     u.SubscriptionStatus,
110     u.TotalWatchTime,
111     u.Device
112 FROM users u
113 LEFT JOIN ratings r
114     ON u.UserID = r.UserID
115 WHERE r.UserID IS NULL;
```

Result Grid Filter Rows: Export: Wrap Cell Content: Fetch rows:

	UserID	Age	Gender	Country	SubscriptionStatus	TotalWatchTime	Device
▶	SF_1	11	F	India	Free	231	Mobile
	SF_7	24	M	Canada	Free	17	Smart TV
	SF_12	31	M	Italy	Free	253	Desktop
	SF_14	14	M	India	Subscriber	465	Tablet
	SF_18	26	F	France	Subscriber	65	Smart TV
	SF_19	32	M	Italy	Subscriber	14	Tablet
	SF_31	18	M	Italy	Free	184	Tablet
	SF_39	39	M	Italy	Free	443	Tablet
	SF_46	24	M	Italy	Subscriber	133	Desktop
	SF_47	35	M	India	Free	278	Smart TV
	SF_51	52	F	USA	Subscriber	471	Mobile
	SF_55	58	F	Canada	Free	479	Desktop
	SF_61	39	M	UK	Subscriber	362	Tablet
	SF_63	59	M	USA	Subscriber	496	Tablet
	SF_65	45	M	Italy	Subscriber	485	Smart TV
	SF_69	57	F	USA	Subscriber	122	Mobile
	SF_74	39	M	Italy	Subscriber	120	Mobile
	SF_82	25	M	France	Subscriber	499	Tablet
	SF_92	62	F	France	Subscriber	392	Desktop
	SF_98	48	F	Italy	Subscriber	336	Smart TV
	SF_101	36	F	India	Free	177	Tablet
	SF_114	38	F	UK	Subscriber	189	Mobile
	SF_125	60	M	Italy	Subscriber	392	Smart TV
	SF_126	16	M	UK	Free	264	Tablet
	SF_128	59	M	UK	Subscriber	186	Desktop
	SF_129	54	M	UK	Subscriber	212	Smart TV
	SF_135	9	M	France	Subscriber	15	Smart TV
	SF_138	59	M	France	Free	348	Smart TV
	SF_147	27	M	UK	Free	379	Smart TV
	SF_149	64	M	Italy	Subscriber	183	Smart TV
	SF_157	42	M	Canada	Free	89	Tablet
	SF_171	28	F	Canada	Free	280	Smart TV
	SF_174	19	M	Italy	Subscriber	70	Smart TV
	SF_175	51	F	UK	Subscriber	478	Tablet

9. Check movies without ratings

SELECT

m.MovieID,

m.Movie_Title,

m.Year,

m.Language,

m.Country,

m.Total_Views

FROM movies_clean m

LEFT JOIN ratings r

ON m.MovieID = r.MovieID

WHERE r.MovieID IS NULL;

The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane displays the 'data' database with a 'datamstreaming' schema containing tables like 'genres', 'movie_genres', 'movies_clean', 'ratings', and 'users'. The 'movies_clean' table is selected. The main editor shows a SQL query:

```
-- 9. Check movies without ratings
SELECT
  m.MovieID,
  m.Movie_Title,
  m.Year,
  m.Language,
  m.Country,
  m.Total_Views
FROM movies_clean m
LEFT JOIN ratings r
  ON m.MovieID = r.MovieID
WHERE r.MovieID IS NULL;
```

 The 'Result Grid' at the bottom displays the query results, showing columns: MovieID, Movie_Title, Year, Language, Country, and Total_Views. The results list 27 movies, including 'Now and Then', 'Richard III', 'How to Make an American Quilt', 'Guardian Angel', 'Confessional, The (Le Confessionnal)', 'White Balloon, The (Badkonake Sefid)', 'Beautiful Girls', 'Bridges of Madison County, The', 'Man of the Year', 'Crimson Tide', 'Feast of July', 'Love & Human Remains', 'Nadja', 'Total Eclipse', 'Death and the Maiden', 'French Kiss', 'Heavenly Creatures', 'Kid in King Arthur's Court, A', 'Pushing Hands', 'Picture Bride', 'Ready to Wear (Pret-A-Porter)', 'Secret of Roan Inish, The', 'Tales from the Hood', 'It Could Happen to You', 'Timecop', 'Beverly Hills, The', 'Bronx Tale, A', 'Cabin Boy', 'Carito's Way', 'Favor, The', 'Heaven & Earth', 'Hot Shots! Part Deux', 'House of the Spirits, The', 'What Happened Was...', and 'Robert A. Heinlein's The Puppet Mast...'. The 'Result 15' tab is active.